

Signalling in the intelligent network

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Intelligent network principles demand flexible, reliable and efficient signalling systems. These qualities are embodied in the intelligent network application protocol and common channel signalling system No 7.

This paper describes the structure and definition of the intelligent network application protocol and its relationship to other signalling system No 7 protocols, including a brief overview of the mobile application part. Use of the intelligent network application protocol is outlined, and future developments are discussed.

1. Introduction

To achieve the goal of rapid service creation and deployment in public switched networks, it is necessary to functionally separate the control of services from basic call processing. The intelligent network (IN) supports this concept, and defines an architecture of modular and reusable network functions that enable service/network independence. Physical distribution of these functions is also supported, giving a network operator the flexibility to build a network using platforms that are specialised to perform specific network functions.

Distribution brings with it the need to communicate. It is essential that the protocols used for signalling between the network functions are service independent, since this allows new services and features to be implemented without upgrades to the signalling systems. Standardisation of the signalling systems is also important, particularly in a multi-vendor network or where interconnect to other networks is required.

In building public telecommunications networks, network operators have developed digital signalling systems that are efficient and resilient. However, many protocols have been shown to be somewhat inflexible, monolithic and specific to particular operators. At the same time, the computer industry has pursued 'open standards' and 'open systems'. Open standards and open systems (systems which conform to the ISO open systems interconnection reference model — OSI) allow distributed processes to communicate and co-operate across one or more networks. Furthermore,

they propose modular and layered approaches, which give the adaptability required in a modern signalling network.

The last few years have seen the maturity of the OSI model, as well as significant developments in computer and software technology. The need for flexibility in network signalling systems has led to the adoption of OSI principles, re-flected in the current international standards for signalling system No 7 and the intelligent network application proto-col.

2. Signalling systems

The software processes in today's integrated digital networks communicate to control the network using common channel signalling system No 7 (SS7) [1—3]. This is a digital, message-based common channel signalling system, primarily for use within the network.

The fact that SS7 is digital and message-based makes it ideal for inter-process communication. Being common channel, it is logically (and can therefore be physically) separated from the bearer circuits to which it relates. This gives a greater degree of flexibility in the signalling network, allowing high functionality, resilience and performance.

Signalling between the network elements in an intelligent network requires the versatility of data communications protocols as well as real time efficiency.

These factors have been the main drive behind the more recent developments of SS7, particularly the need to support:

- circuit related signalling — the transfer of information which relates to the establishment, supervision and release of bearer circuits,
- non-circuit related signalling — the transfer of information which does not relate to the switching of connections. This information transfer is required to provide services (e.g. for communications between the call control processes and the processes controlling the supplementary services), and is primarily transaction based.

Recommendations for SS7 are produced by the telecommunications standardisation sector of the ITU (referred to as ITU-T, formerly CCITT), and are standardised by regional standardisation bodies, such as the European Telecommunications Standards Institute (ETSI).

Currently, the majority of network users have analogue access to the network. Signalling between the user's equipment (e.g. telephone) and the network is based on on-hook/off-hook detection at the local exchange, and pulsed or DTMF (dual tone multifrequency) dialling within the speech band (bearer circuit). In many cases, digital access is also supported, usually as an ISDN user-network interface (UNI). This supports common channel, digital, message based signalling, referred to as Digital Subscriber Signalling System No. 1 (DSS1) [4]. Recommendations for DSS1 are again produced by the ITU-T and standardised by the regional standardisation bodies.

3. Protocol reference models

The evolution of the ITU-T digital signalling systems has followed the development of the ISO open systems interconnection (OSI) reference model [5]. The OSI protocol reference model defines a seven-layer logical structure for the development of protocols used for communications between open systems. For each layer in the model, the purpose of the layer, the services it offers, and the services it requires from the layer below are described. Routeing and addressing within the structure are also described.

The specification of SS7 predates the OSI model. While it does have a layered structure, it consists of only four levels. However, an important aspect of the SS7 protocol reference model (shown in Fig 1) is that there is a clear separation of the application process (user part) from the underlying protocols (message transfer part — MTP — which is common to all of the user parts). The function of the message transfer part is to provide a reliable signalling connection between any two points in the network. To do this, it comprises three levels:

- signalling data link (MTP level 1) — provides a physical bi-directional transmission path between two directly connected network elements,
- signalling link (MTP level 2) — provides a reliable logical signalling channel between two directly connected network elements,
- signalling network (MTP level 3) — provides routeing of signalling messages from any point in the SS7 signalling network to any other point. The routeing of messages is based on the MTP point code, which uniquely identifies an SS7 signalling terminal in the network, and the service information octet, which identifies the user part, or MTP level 3 at that signalling terminal. The MTP level 3, also provides for traffic, route and link management of the signalling network, for example automatically re-routeing a signalling connection on detection failure at level 2.

A number of SS7 user parts have been defined by the ITU-T. These include the telephony user part (TUP), to support basic telephony, and the ISDN user part (ISUP), to support ISDN call and connection control. To support early implementation of SS7 in the BT Network, BT developed an enhanced version of the TUP, known as the BT national user part (BT NUP). This user part supports basic telephony and some ISDN features. Each of these user parts is specified as a single layer structure.

The signalling connection control part (SCCP) was added by the ITU-T as a user part to the SS7 protocol reference model to align SS7 with the OSI model. The SCCP provides an OSI network layer conformant interface to the MTP, enabling OSI conformant higher layers to be used with SS7. The SCCP provides routeing of signalling messages based on a global title (a globally valid address) which it translates to an MTP point code and SCCP subsystem (SCCP user) number. It also supports connectionless and connection-oriented services, and segmentation and re-assembly of higher-layer signalling messages. The combination of the MTP and SCCP is referred to as the network services part (NSP), which provides reliable transfer and global routeing and addressing of signalling messages through the network. Users of the SCCP are the ISDN user part (for some end-to-end signalling procedures) and the SS7 transaction capabilities.

To date, no protocols have been specified to support the transport, session and presentation layers of the OSI model in SS7. These layers are referred to as the intermediate services part (ISP), and are responsible for the end-to-end transfer and presentation of information to and from the application layer.

The transaction capabilities of SS7 are designed to interface directly to the OSI network layer, provided by SCCP in SS7. Currently, the only transaction capabilities

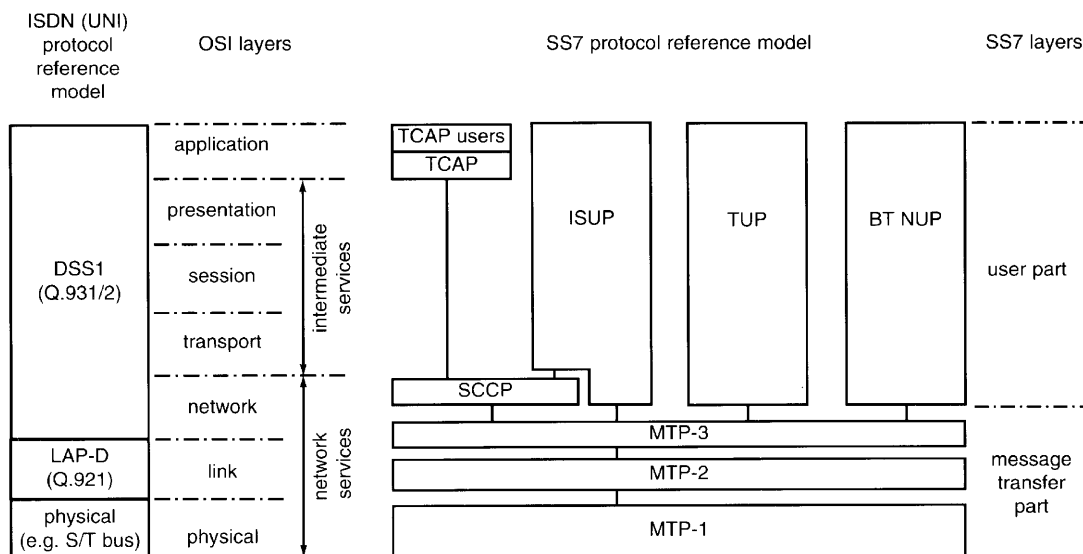


Fig 1 SS7 and ISDN (UNI) protocol reference models.

are defined in the transaction capabilities application part of SS7 (TCAP). This is an application layer protocol that uses the SCCP connectionless service, and does not require an ISP. This makes the signalling system efficient in that TCAP messages do not have to be processed by ISP layers, but loses the flexibility that those layers provide.

Other application layer protocols have been defined for use with SS7, and use the services provided by TCAP. These protocols, which include the intelligent network application protocol (INAP) and the mobile application part (MAP), are referred to as TCAP users.

4. Signalling in the intelligent network

A feature of the intelligent network architecture is that it allows the platforms providing basic services (call processing) to be physically separate from the platforms providing supplementary services. Non-circuit related signalling is required to convey information between the applications running on these platforms.

The approach taken to define the signalling for the new interfaces defined in the IN was to have a clear separation between the application-specific protocol and the underlying protocols that support it. This enables the application-specific protocol to be designed for high functionality and flexibility, and the underlying protocols designed for efficiency and resilience.

The network services part of SS7 (MTP and SCCP) fulfils the requirements for the underlying protocols; the OSI application layer (TCAP and TCAP users in the SS7 protocol reference model) fulfils those of the application specific protocol. The TCAP user for IN signalling is the intelligent network application protocol (INAP). The INAP

design methodology and protocol are described in the following section.

5. The intelligent network application protocol

5.1 The IN conceptual model

The IN conceptual model [6] defines four planes of decreasing abstraction — service, global functional, distributed functional and physical. These planes are used as the basis for the specification and design of the intelligent network application protocol to support services in an IN structured network.

At the service plane, a target set of services to be supported are identified and described in an implementation independent way, and decomposed into service features. The service features are realised in the global functional plane, the next level down, by global service logic. The global functional plane has an overall abstract view of the whole network and its capabilities. These capabilities are modelled as service-independent building blocks (SIBs), which are used by the global service logic to build services.

A number of SIBs are defined in IN capability set 1 (CS-1), covering areas such as charging, user interaction and call queuing. The basic call-processing capability of the network is abstracted as a single SIB, called the basic call process.

Functional entities within the network are identified at the distributed functional plane. This allows the functionality of the global service logic and SIBs to be distributed across the functional entities. In performing this distribution, functional relationships between the functional

entities are defined. It is these functional relationships which determine the requirements of the signalling protocols used in the IN.

5.2 Information flows in the distributed functional plane

The distributed functional plane for IN CS-1 [7] defines a functional architecture comprising a number of functional entities and functional relationships. Part of this functional architecture is shown in Fig 2. The functional entities are detailed further in Abernethy et al [6]. In the distributed functional plane, each of the SIBs are broken down into actions to be performed by the functional entities, and information that flows between them.

Each of the functional relationships between the functional entities (denoted in Fig 2 by the reference points A to F) require control capabilities. These are:

- connection control (reference points A, B, C) — to establish, supervise and release a bearer connection,
- call control (reference points A, B) — to initiate and terminate the call, and provide the end-to-end control required for supplementary services which do not require IN service control,
- IN service control (reference points D, E, F) — to provide the control capabilities required for IN supplementary service control.

The distributed functional-plane architecture uses the call-control agent and call-control functional entities and the functional relationships (A,B) defined for these ISDN functional architecture [8]. These functional entities provide the non-IN call/connection control capabilities, referred to as basic call processing. This is modelled in the distributed functional plane as the basic call state model [6].

The only time that the IN service control interacts with the basic call state model is when armed detection points are reached in the call. When this occurs, information is passed

to the service-switching function, which screens the information against the detection point criteria, e.g. whether the dialled digits match a particular digit sequence. If the criteria are met, then the service-switching function will initiate service control.

The non-IN call/connection control information flows (at the A and B reference points) have been defined as part of the ISDN, and form the basis of the ISDN signalling protocols. The information flows at the D, E and F reference points are defined as part of IN CS-1. The functional relationship between the call control and specialised resource functional entities (the C reference point) is considered to have only the connection-control information flows defined as for the A and B reference points. Due to the close coupling of the call-control and service-switching functional entities, the information flows at the M reference point are not standardised in IN CS-1.

The information flows at the D, E and F reference points are identified for each SIB. Figure 3 gives an example of some of the information flows and actions that make up the user interaction SIB. This SIB is used in cases where user interaction with a party is required in a service, e.g. to play an announcement or collect DTMF tones. In the example in Fig 3, the SCF requires the CCF/SSF to connect the user to a specialised resource, and then instructs the specialised resource to play an announcement to the user. Once the announcement has been played — indicated by a report from the SRF — the bearer connection between the CCF/SSF and the SRF is released, unless further interaction is needed.

The complete set of information flows between any two functional entities, needed to support all of the declared SIBs, defines the functional relationship between those two entities. Some of the information flows are common to more than one SIB. For example, the connect to resource information flow is common to both the user interaction and queue SIBs.

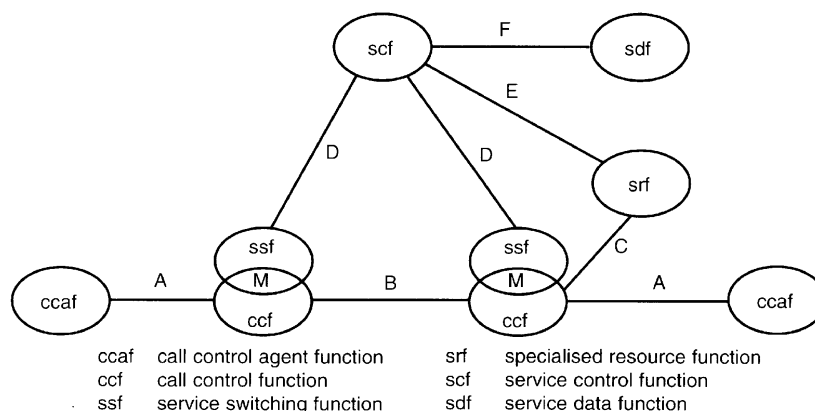


Fig 2 Call control and service associated functional entities.

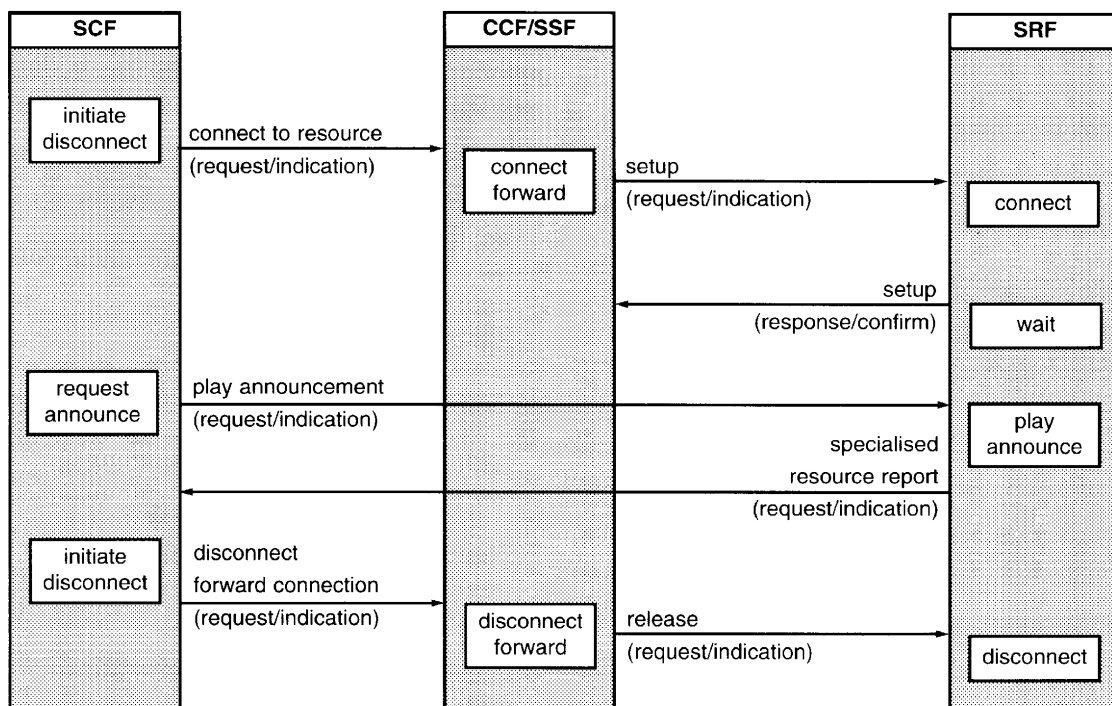


Fig 3 Example functional entity actions and information flows in the user interaction SIB.

5.3 The physical plane

The capabilities of the IN are realised in the physical plane of the IN conceptual model. Each of the functional entities are assigned to physical network elements, for example, as shown in Fig 4 [6].

In some physical implementations, different functional entities may reside within the same physical entity (as in the SCP shown in Fig 4). In these cases, the relationship between the functional entities is internal, and is not standardised. However, the INAP supports all of the functional relationships (*D, E* and *F*) as physical interfaces.

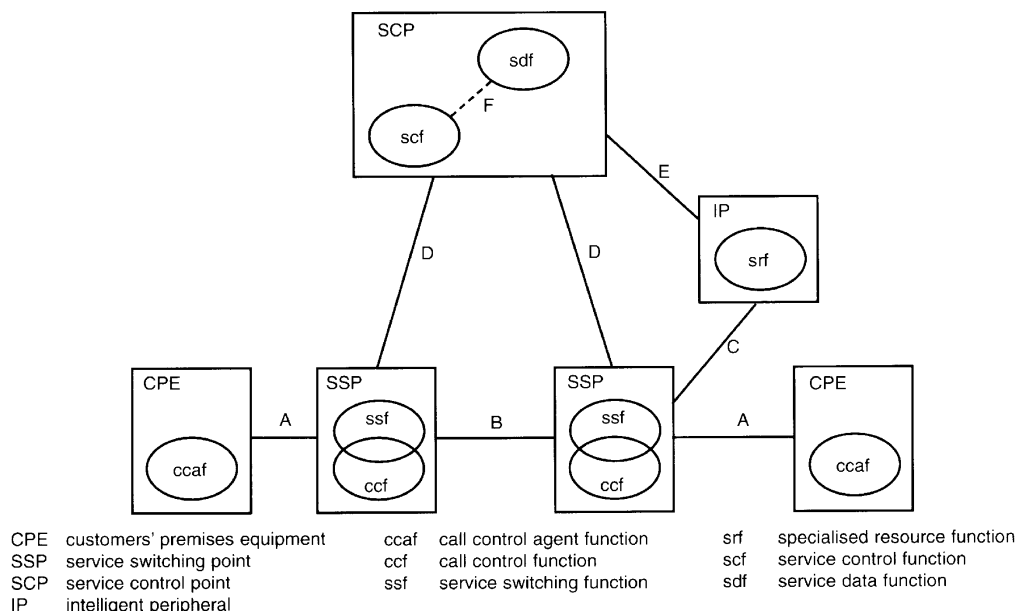


Fig 4 Example physical plane showing allocation of functional entities to physical entities.

5.4 INAP protocol structure

The ITU-T recommendation Q.1400 [9] covers the development of signalling protocols based on the OSI application layer structure. This is a modular structure which enables the application layer to provide specific, expandable communications services to an application, while using generic underlying protocols. This structure has been adopted for the INAP, as it is ideally suited to the computer/computer communications required in the IN.

In the OSI environment, application processes (such as the SCF and SSF in the IN) contain communications application entities (AEs). These AEs are essentially the protocols and mechanisms that are responsible for all of the communications between peer applications. If an application process contains more than one application entity, it is the responsibility of the application process to control and co-ordinate them. For example, a service control function will contain different application entities to communicate with the service management function (typically a management oriented protocol) and the service switching function (typically INAP).

An actual instance of an application process, i.e. running software, is referred to as an application process invocation (API). Each API will contain one or more application entity invocations (AEIs). This is illustrated in Fig 5.

The functional relationships between the applications are represented by application associations between an application entity and its peers. For each association, a single association object (SAO) exists within the AE which represents all of the communications capabilities required for that application association. These communications capabilities include the state models, operations, result reporting and error handling needed to support an association.

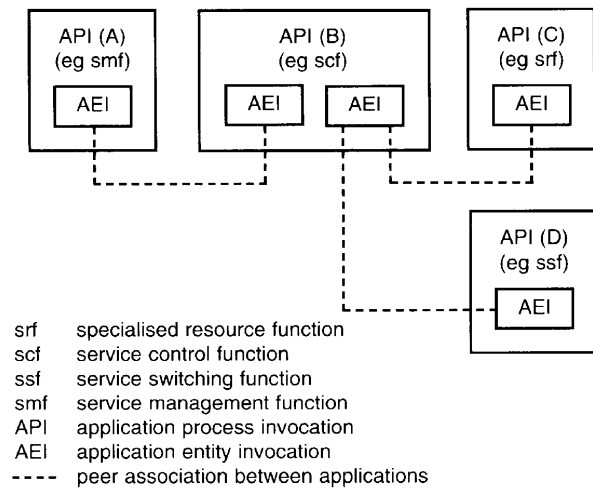


Fig 5 Relationship between APIs and EAIs.

The communications capabilities are defined in related groups, referred to as application service elements (ASEs). The ASEs are governed within the SAO by a single association control function (SACF), which governs the necessary ordering and co-ordination of the ASEs. If more than one SAO exists within an application entity, the multiple application associations are governed by a multiple association control function (MACF). The structure of application entity invocations with one and multiple SAOs is shown in Fig 6.

Not all of the ASEs within an SAO may be needed for a particular application association. An application context defines which ASEs and any related options that are required to support a specific application association, and is

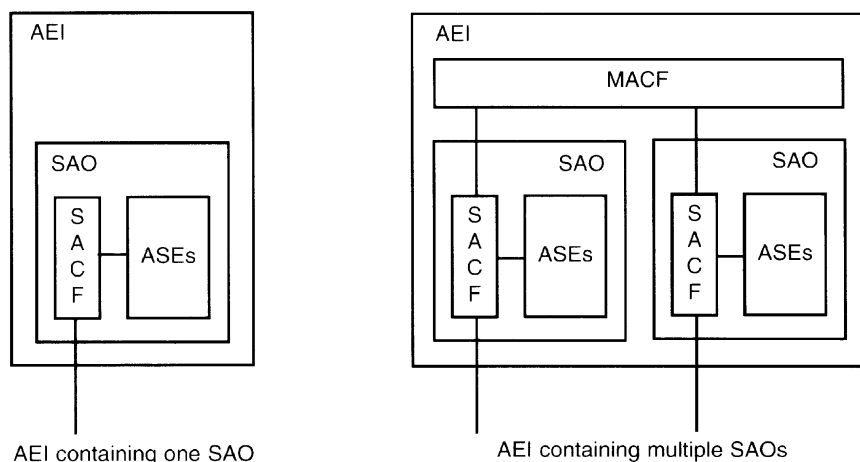


Fig 6 Application entity invocation structures.

agreed by both peer application entities prior to an association being initiated.

The SS7 TCAP is an ASE which provides a framework for other ASEs and mechanisms for controlling an application association. It comprises two sublayers — the transaction sublayer and the component sublayer. The TCAP transaction sublayer is based on the OSI association control service element (ACSE [10]) and is responsible for establishing and releasing an application association between two application entities. Establishment includes defining the application context for the association and must therefore occur before any information is sent or received by other ASEs in the SAO.

The TCAP component sublayer is based on the OSI remote operations service element (ROSE) [11]. This is a framework for defining remote operations in a distributed open systems environment. A remote operation is an operation which an application entity (invoker) requests another application entity (performer) to perform. The performer replies with the outcome of performing the operation, either a valid result or an error.

Remote operations supported by ROSE are classed by whether they are asynchronous or synchronous (i.e. whether the invoker may invoke further operations before the reply from the first is received or not), and whether success and/or failure or neither are reported. Linking together of remote operations may occur such that the performer of an initial operation may invoke further operations on the invoker of

that operation, before the reply to the initial operation is returned.

These remote operations supported by ROSE must be invoked within an application association. They are supported in TCAP as simple structured or unstructured dialogues. Some enhancements to the ROSE protocol are included in TCAP, although only asynchronous remote operations are supported.

Figure 7 shows the structure of a service control function application process invocation in an SS7 environment.

5.5 INAP protocol definition

In the physical plane of the IN conceptual model, the information flows identified in the distributed functional plane are defined as remote operations consistent with the ROSE framework. The remote operations have optional arguments (corresponding to information elements), optional responses (results and errors), and optional linked operations which can be used as the response to a prior operation request.

The remote operations are grouped to form ASEs that define which entities invoke which operations. A number of ASEs relate to each physical interface, or SAO. The intelligent network application protocol is the collection of all of the ASEs and the necessary SACF and MACF rules, together with the definition of the actions taken at each entity to invoke and perform the required operations.

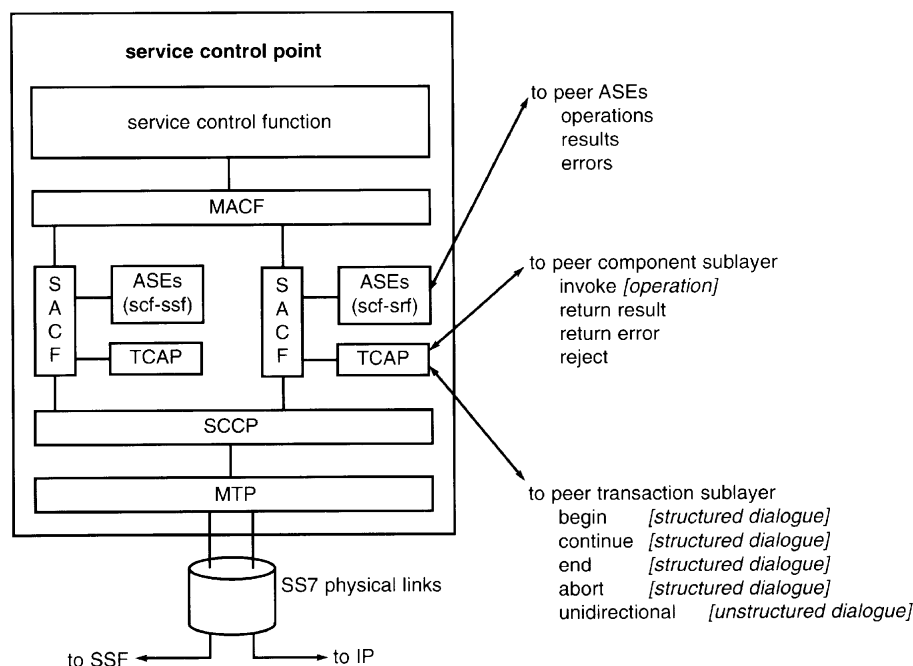


Fig 7 SS7/INAP protocol model (at service control point).

Since INAP is a ROSE user protocol it can be supported by any protocol stack that supports ROSE. This includes SS7 (TCAP) and DSS1. The INAP remote operations are defined in abstract syntax (ASN.1) [12], which makes them implementation-independent and portable to any processing environment.

Figure 8 shows an example of some of the operations between the SCF and SRF (which realise two of the information flows shown in Fig 3). The 'play announcement' operation has an argument (PlayAnnouncementArg) and no defined results. This operation can be invoked by an SCF and is performed by an SRF. The argument will contain operation specific information, such as identifying the announcement to play. In this example, two errors are possible (Cancelled and MissingParameter). These could occur if the operation is subsequently cancelled by the SCF, or if the SCF does not supply the correct arguments when invoking the operation. Since the 'play announcement' operation does not return any results, a linked operation (SpecialisedResourceReport) is identified which can be used to return any information to the SCF about the execution of the 'play announcement' operation.

```

PlayAnnouncement                               ::= OPERATION
  ARGUMENT
    PlayAnnouncementArg
  ERRORS {
    Cancelled,
    MissingParameter
  }
  LINKED {
    SpecialisedResourceReport
  }

PromptAndCollectUserInformation                 ::= OPERATION
  ARGUMENT
    PromptAndCollectUserInformationArg
  RESULT
    ReceivedInformationArg
  ERRORS {
    Cancelled,
    ImproperCallerResponse
  }

SpecialisedResourceReport                      ::= OPERATION
  ARGUMENT
    SpecialisedResourceReportArg

```

Fig 8 Example abstract syntax definitions of some SCF-SRF operations.

The 'prompt and collect user information' operation has an argument, result and two possible errors. This operation instructs the SRF to prompt the user to enter information, which is collected by the SRF, and returned to the SCF as the result of the operation. No linked operations are specified. The final operation, the 'specialised resource report', only has an argument specified (SpecialisedResourceReportArg), and is used to pass information (such as status reports) from the SRF to the SCF.

Figure 9 gives an example of an ASE to support specialised resource control. In this example, the ASE

contains the three operations shown in Fig 8. It defines that the supplier (in this case the SCF) invokes the 'play announcement' and 'prompt and collect user information' operations, and the consumer (the SRF) invokes the 'specialised resource report'.

```

Specialised-resource-control-ASE ::= APPLICATION-SERVICE-
ELEMENT

-- consumer is SSF/SRF
CONSUMER INVOKES {
  specialisedResourceReport
}

-- supplier is SCF
SUPPLIER INVOKES {
  playAnnouncement,
  promptAndCollectUserInformation
}

```

Fig 9 Example INAP ASE (specialised resource control).

The INAP operations defined in abstract syntax are encoded into a concrete syntax (i.e. bit patterns) using the basic encoding rules [13]. The result is a set of encoded TCAP components which can be transferred to peer entities within a TCAP dialogue.

TCAP component sublayer primitives allow the encoded INAP operations to be passed to TCAP for transmission, as well as providing an interface to control dialogues. The information relating to the operations, such as the name of the operation and arguments are carried in the component portion of a TCAP message. Information relating to the dialogue is transferred in the transaction portion. An optional dialogue portion can be used by the TCAP component sublayer to specify and negotiate application contexts.

When supporting the INAP, the TCAP component sublayer communicates with its peer using the INVOKE, RETURN RESULT (last), RETURN ERROR and REJECT component types. These component types are carried within the component portion of a TCAP message. The transaction sublayer establishes and maintains a structured dialogue with its peer using the BEGIN, CONTINUE, END and ABORT message types, or supports an unstructured dialogue using the UNIDIRECTIONAL message type. These are carried in the transaction portion of a TCAP message. An example of message transfers between an SCP and an intelligent peripheral (IP), to request the IP to prompt a user to enter information (e.g. dial digits) and return them to the SCP, is shown in Fig 10.

The ITU-T has defined the full set of ASEs required to support IN CS-1 [14]. In Europe, ETSI have decided to standardise a core set of these ASEs, referred to as core-INAP [15]. Application contexts are standardised for each of the functional interfaces and each of the various standards. The relationship between the application contexts, ASEs and operations defined in core-INAP is shown in Fig 11.

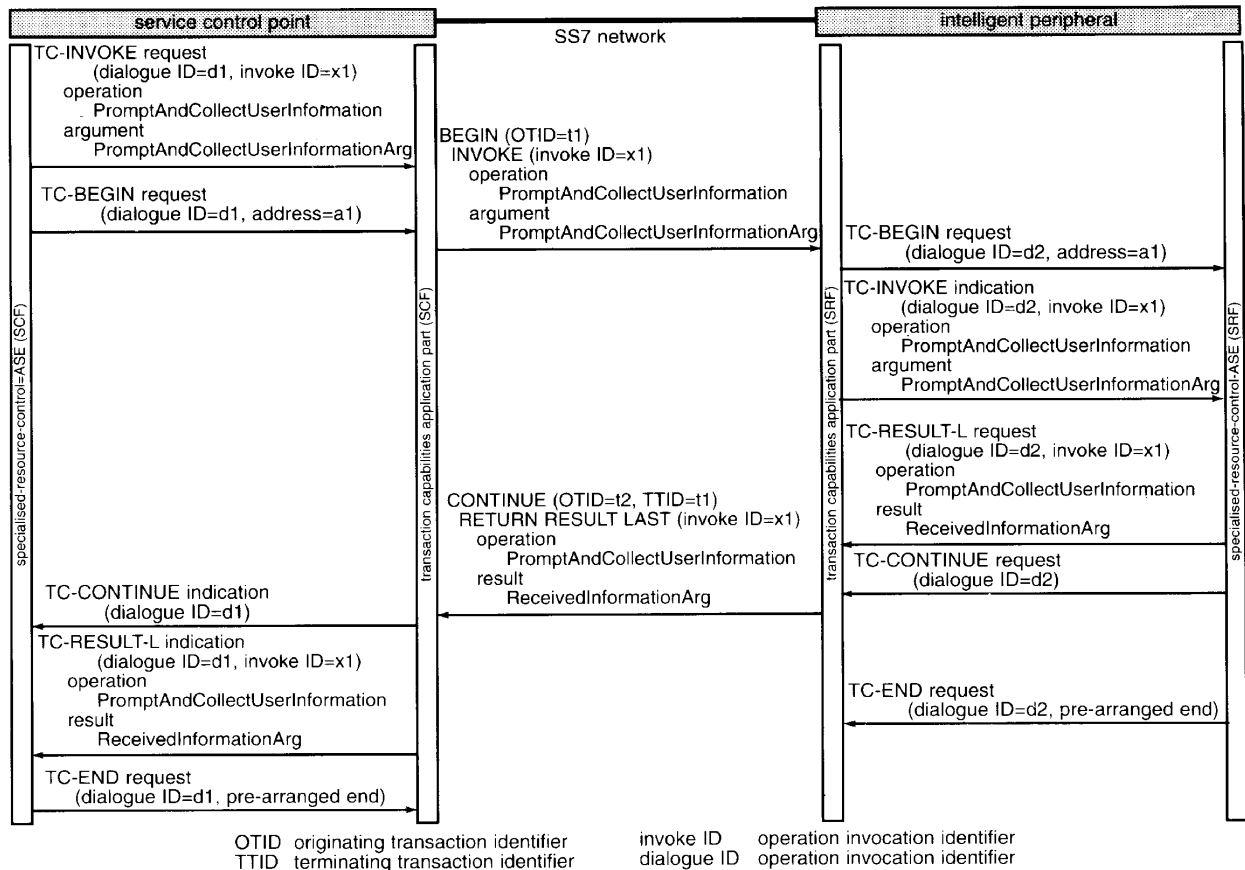


Fig 10 Example message sequences between an SCP and IP.

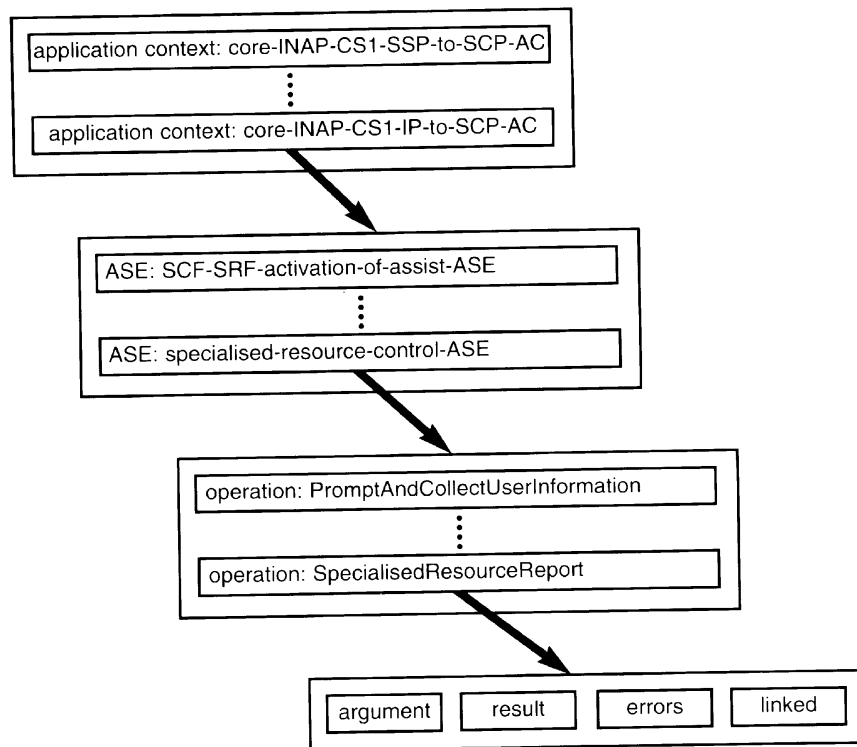


Fig 11 Relationship between application contexts, ASEs and operations.

Addressing of the INAP application entities is performed using the SCCP addressing mechanisms (global title and sub-system number). Each application entity is identified by a global title, which is mapped to an SCCP sub-system number and MTP point code for routing through the SS7 signalling network. Dialogues between application entities are identified in TCAP by a dialogue identifier. The ASEs required for the application association are identified by the application context.

5.6 INAP application entity procedures

INAP specifies the application entity procedures that relate to each of the INAP interfaces. These are specified as finite state models which define the procedures for using the SACF, MACF (if present), INAP ASEs (not including TCAP) and some of the functions of the application process invocation. The finite state models defined in the ETSI core-INAP contain a number of finite state machines.

- A functional entity access manager manages the sending, receiving, queuing and formatting of messages to and from the application entity finite state model.
- One (or more) management entity maintains dialogues with other application entities and performing operations which relate to management or supervisory functions (such as call gapping in the SSF). If more than one management entity is necessary, a management entity control finite state machine is specified. This is responsible for creating and terminating management entities and the call associated finite state machines, and interfacing these to the functional entity access manager.
- One (or more) finite state machine performs the operations associated with single calls. These interface to the functional entity access manager via the management entity control finite state machine or the single management entity.

The structure of the SCF finite state model is shown in Fig 12.

6. Signalling in the BT intelligent network

BT intends to support the ETSI core INAP signalling in its network. The ability to support INAP has required a major upgrade to the signalling network, which is currently under way. The upgrade will enable SS7 non-circuit-related signalling, using SCCP and TCAP, as well as introducing the ISUP protocol for ISDN call control. The upgraded network is shown in Fig 13.

The use of SCCP and TCAP are primarily to support the ETSI core INAP protocol between SSPs, SCPs and IPs. To enable all SSPs (local and transit exchanges) and IPs to be

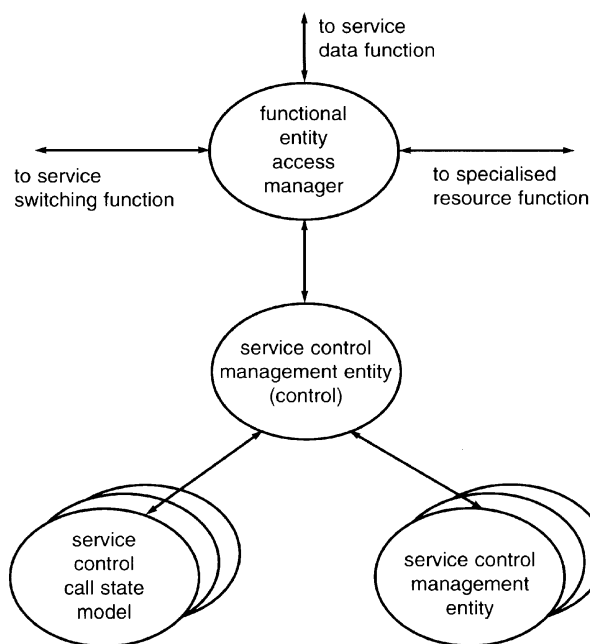


Fig 12 The structure of the SCF application entity finite state model (ETSI Core INAP).

connected efficiently to all SCPs, signalling point relays have been introduced into the SS7 non-circuit-related signalling network. These provide global title routing of SCCP messages between the SSPs/IPs and the SCPs, and sub-system management. This not only reduces the number of connections required, but also increases the resilience of the network by providing routing to an alternative (e.g. stand-by) SCP in the case of an SCP failure.

The BT network currently supports both analogue in-band access signalling (for example, pulsed and DTMF dialling) and ISDN access signalling (DSS1 and its own digital access signalling system No. 2). Some of the IN-based services, such as number translation services, can be provided via all of these signalling systems. Others, particularly those which require user interaction, require the user to send DTMF tones. These services cannot be accessed from an ISDN terminal unless it is equipped with a DTMF tone generator. Services which currently use voice announcements, such as calling card services, obviously cannot be used with ISDN data calls.

7. Signalling in the mobile intelligent network

Recommendations for signalling in public land mobile networks were published by the CCITT in 1988 [16], and were an attempt to define a signalling protocol which could support mobile facilities for voice and non-voice services. The mobile application part (MAP) of SS7 was

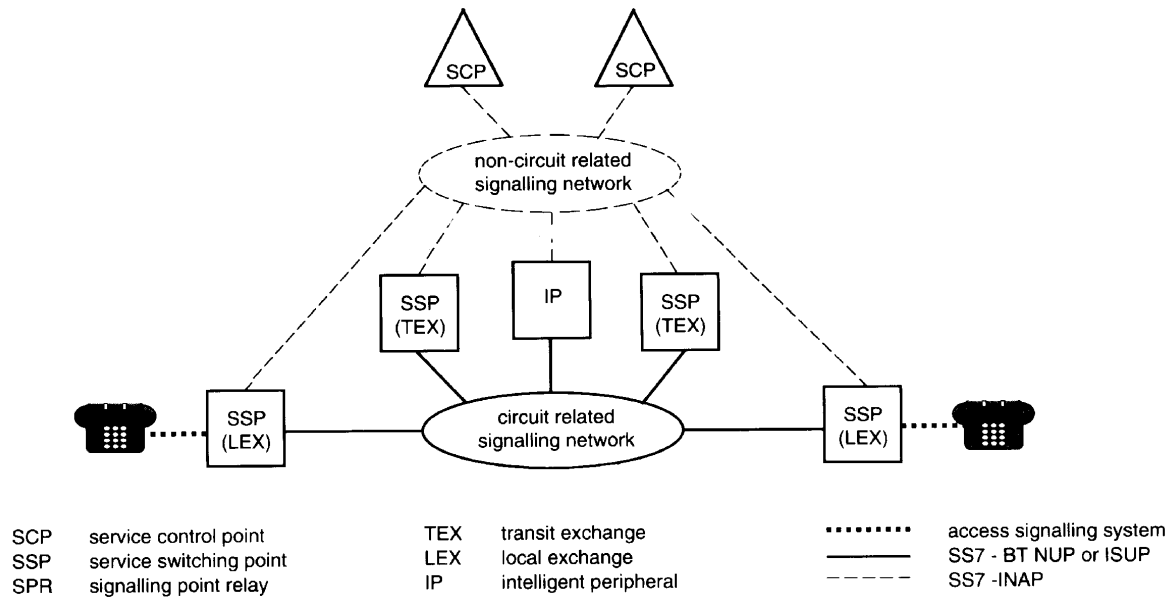


Fig 13 The BT signalling network.

developed based on the public land mobile network architecture and the procedures identified to support mobility (mainly roaming and handover).

In Europe, ETSI have standardised the global system for mobile communications (GSM) network, based upon the public land mobile network [17]. This uses the MAP protocol for non-circuit-related signalling within GSM networks and interconnect between GSM networks. The MAP protocol has been further developed by ETSI, both in functionality and the interfaces supported. The GSM signalling network architecture is shown in Fig 14.

In the GSM network, signalling between the mobile station and mobile switching centre (MSC) uses the GSM radio interface layer 3 (RIL3) protocols. These protocols are based on ISDN access signalling (DSS1) and support the call control, radio resource management and mobility management required by mobile terminals. Between the mobile station and base station sub-system (the air interface), the RIL3 protocols are supported by a radio resource protocol. Between the base station sub-system and the MSC, the RIL3 protocols are supported on SS7 by the direct transfer application part. Control of the base station sub-system from the MSC is performed by the base station sub-system management part (BSSMAP) of SS7.

SS7 is used within the network for circuit-related and non-circuit-related signalling. Circuit-related signalling is required between MSCs and between MSCs and gateways to other networks. This can be provided by SS7 ISUP or operator specific protocols, such as BT NUP.

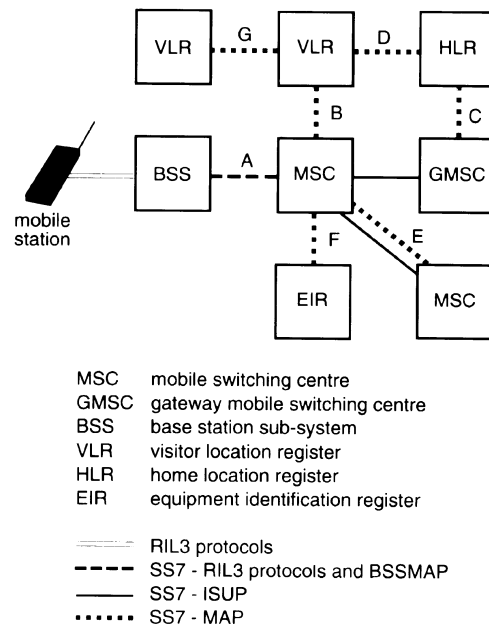


Fig 14 The GSM network signalling architecture.
(Note: the letters A-G indicate the functional interfaces. The H & I interfaces are not shown.)

Non-circuit-related signalling is required between MSCs, home location registers, visitor location registers and equipment identification registers and is provided by SS7 MAP. MAP interfaces are also specified between the mobile station and home location register (I interface) for invoking services, and between the MSC and a messaging network gateway (H interface), to support the GSM short-message service.

The MAP protocol is specified as a ROSE user protocol, and has operations grouped into ASEs to support each of the identified functional relationships. However, MAP does conform to the application layer structure for signalling systems (ITU-T Q.1400), since this was not established at the time MAP was developed. The SS7 protocol model used for MAP is shown in Fig 15.

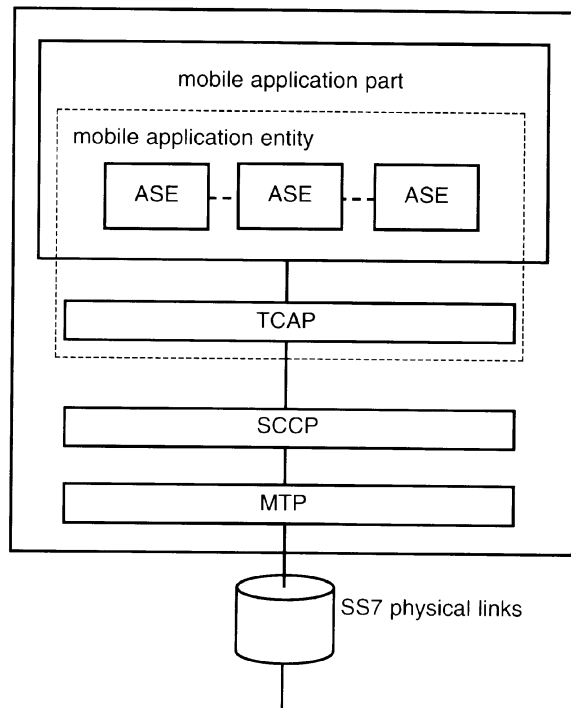


Fig 15 The SS7 MAP protocol model.

No further development of the MAP protocol has occurred in the ITU since the initial 1988 recommendation. It is expected that any future network protocols to support mobility will be part of the INAP.

8. Future developments

The use of the OSI application layer structure as a framework for the development of the INAP has resulted in an adaptable protocol that is well placed to take advantage of current and future trends in computing and telecommunications. As a ROSE user, the INAP can be used with any communications system that supports ROSE functionality, including current signalling systems such as SS7 and DSS1. Having a modular structure enables the INAP to be easily enhanced to support new capabilities and interfaces.

The next phases of IN standardisation, capability sets 2 and 3, will place new demands on the INAP protocol. Targeted services — including universal personal telecommunications (UPT), global virtual network communications (GVNS) and broadband ISDN (B-ISDN) — will require new capabilities and new interfaces (e.g. SCP to SCP) to be supported within networks and across network boundaries. This may include the need to support INAP in the access

network, for example, to support UPT registration from ISDN or mobile terminals.

It is expected that co-operation between the network and intelligent CPE in providing communications services will be a strong driver in the development of future intelligent networks. Extension of the INAP to the user via the ISDN access signalling channel (D-Channel) would increase the user's ability to interact and control the network. Other aspects of ISDN, such as the use of service and terminal capability information to determine what type of basic service is being provided and how, also require further consideration.

The use of the intelligent network is applicable to B-ISDN, although alignment of IN and B-ISDN is not expected until IN CS-3. Broadband ISDN network signalling [18] has also been developed in accordance with the OSI application layer structure. However, while the B-ISDN functional architecture does not include service control, a functional separation between the call control functionality (end-to-end signalling) and the bearer control functionality (required to switch and control the bearer circuits) has been made.

This separation needs to be taken into account by the IN conceptual model, and will impact on the INAP, e.g. new information elements will be required in INAP operations to support connection bandwidth and quality of service. New detection points in the basic call state model and service switching functionality will need to be defined (e.g. to support number translation on terminal-to-terminal look-ahead messages, requests to modify bandwidth, or to control multi-party, multi-connection calls) which could result in new INAP operations and ASEs.

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